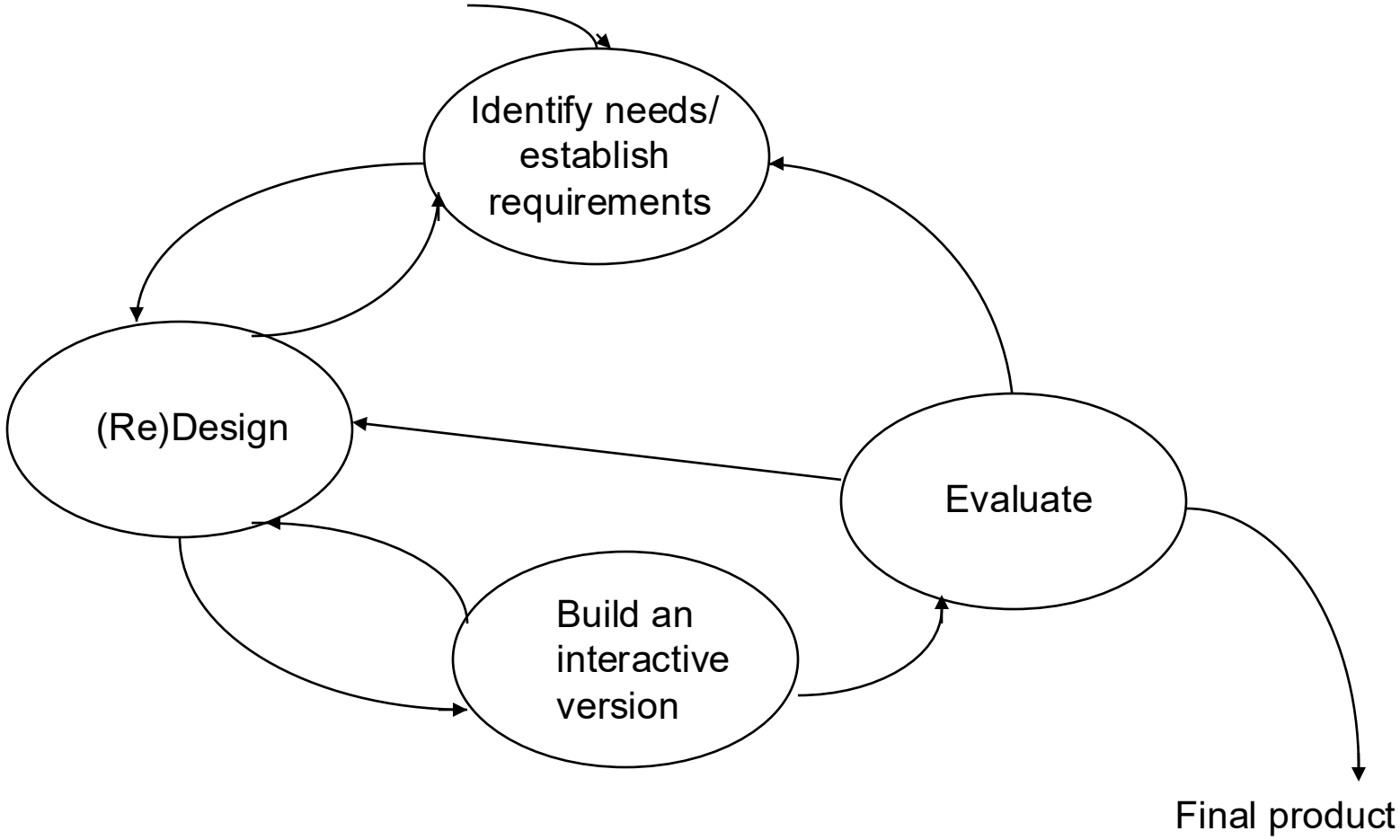
Human Computer Interaction
Week 7

Dr. Sevgi Koyuncu Tunç

User Interface Design & Prototyping

A model for interaction design



What is User-Centered Design?

- An approach to UI development and system development.
- Focuses on understanding:
  - Users, and
  - Their goals and tasks, and
  - The environment (physical, organizational, social)
- Pay attention to these throughout development

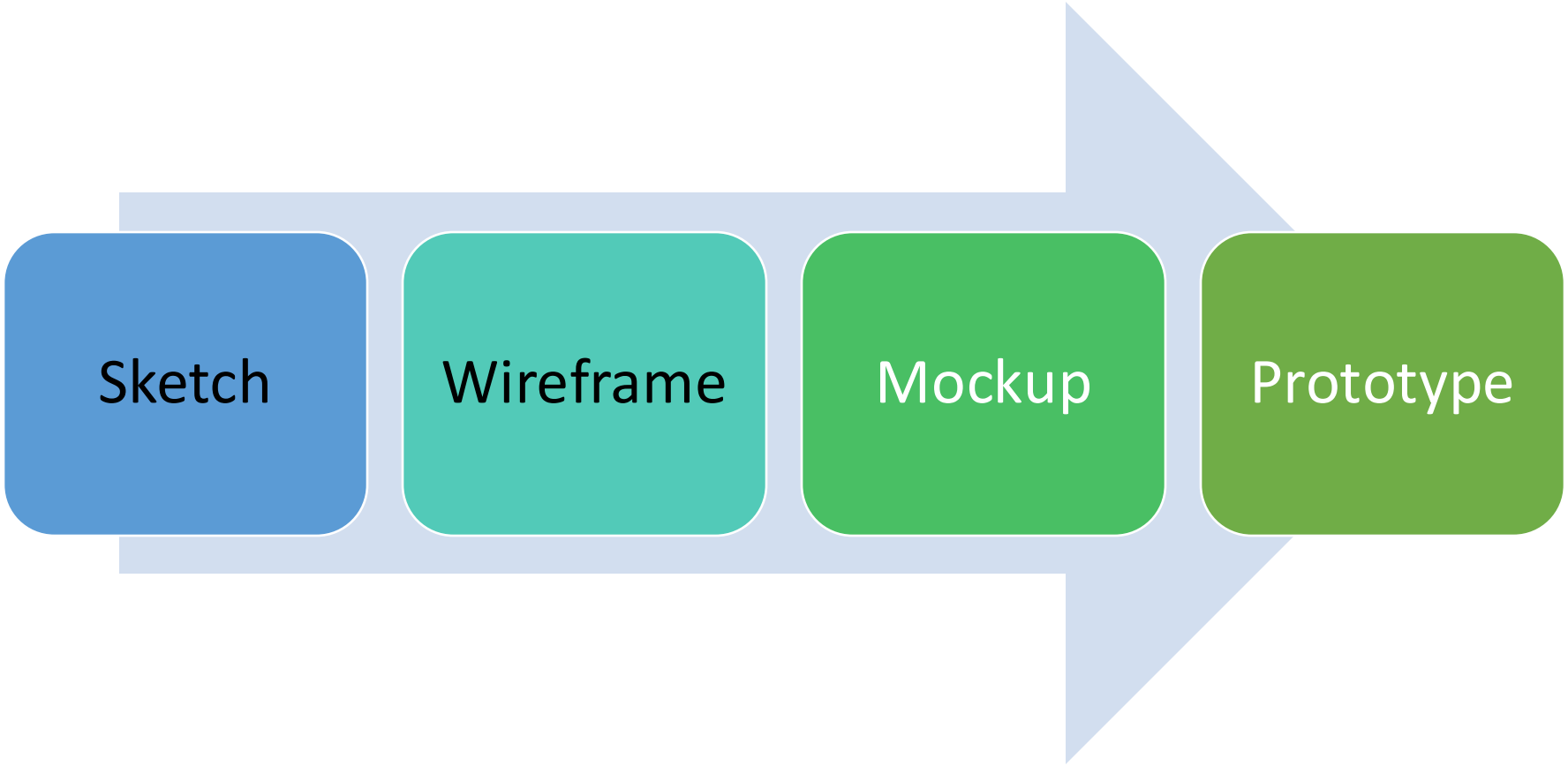
ISO on User-centered Design

- ISO 13407 describes human-centered design processes for interactive systems
- Principles of human-centered design:
  - Active involvement of users
  - Appropriate allocation of function between user and system
  - Iteration of design solutions
  - Multidisciplinary design teams

ISO on User-centered Design (2)

- Essential activities in human-centered design:
  - Understand and specify the context of use
  - Specify the user and organizational requirements
  - Produce design solutions (prototypes)
  - Evaluate designs with users against requirements

User Interface Design



What is a prototype?

- What do you think of when you hear “prototype”?
- What kinds of prototypes have you seen anywhere?
  - in other fields or disciplines?
  - on television?
- What are they “for”?

What is a prototype?

- In other design fields a prototype is a small-scale model:
  a miniature car
  a miniature building or town
- Exists for some purpose
  - Show the “concept” to some stakeholders
  - Get feedback about some aspect
  - Test somehow
    - E.g. a wing in a wind-tunnel

Prototyping and Software

- Do software companies do this?
  - Sometimes do it well
  - But sometimes the prototype is…

    Version 1.0!

- Constantine and Lockwood:

  "Software is the only engineering field that throws together prototypes and then attempts to sell them as delivered goods."

What is a prototype for us?

In HCI / interaction design it can be (among other things):

- a series of screen sketches
- a storyboard, i.e. a cartoon-like series of scenes
- a Powerpoint slide show
- a video simulating the use of a system
- a cardboard mock-up
- a piece of software with limited functionality written in the target language or in another language

Why prototype in general?

- Evaluation and feedback are central to interaction design
- Developers can test feasibility of ideas with team, users
- Stakeholders can see, hold, interact with a prototype more easily than a document or a drawing
- Team members and users can communicate effectively
- To validate existing / other requirements
- It encourages reflection: very important aspect of design
- Prototypes answer questions, and support designers in choosing between alternatives

What to Prototype and Why

- Prototyping reduces uncertainty
  - It can be a major tool for ***risk management***
  - Apply on whatever you might be uncertain about!

- Prototyping technical issues
  - E.g. run-time issues
- Prototyping to establish requirements
  - Users “see” functionality
- Prototyping for usability concerns
  - Our concern in this course

What we need to start UI design?

- You need to understand the following info to be able to design the elements and plan the actions.
- ✅ **Data structure**
- ✅ **Business Rules/Conditions/Validations:**
- ✅ **Actions**
- ✅ **Links**

Low-fidelity Prototyping

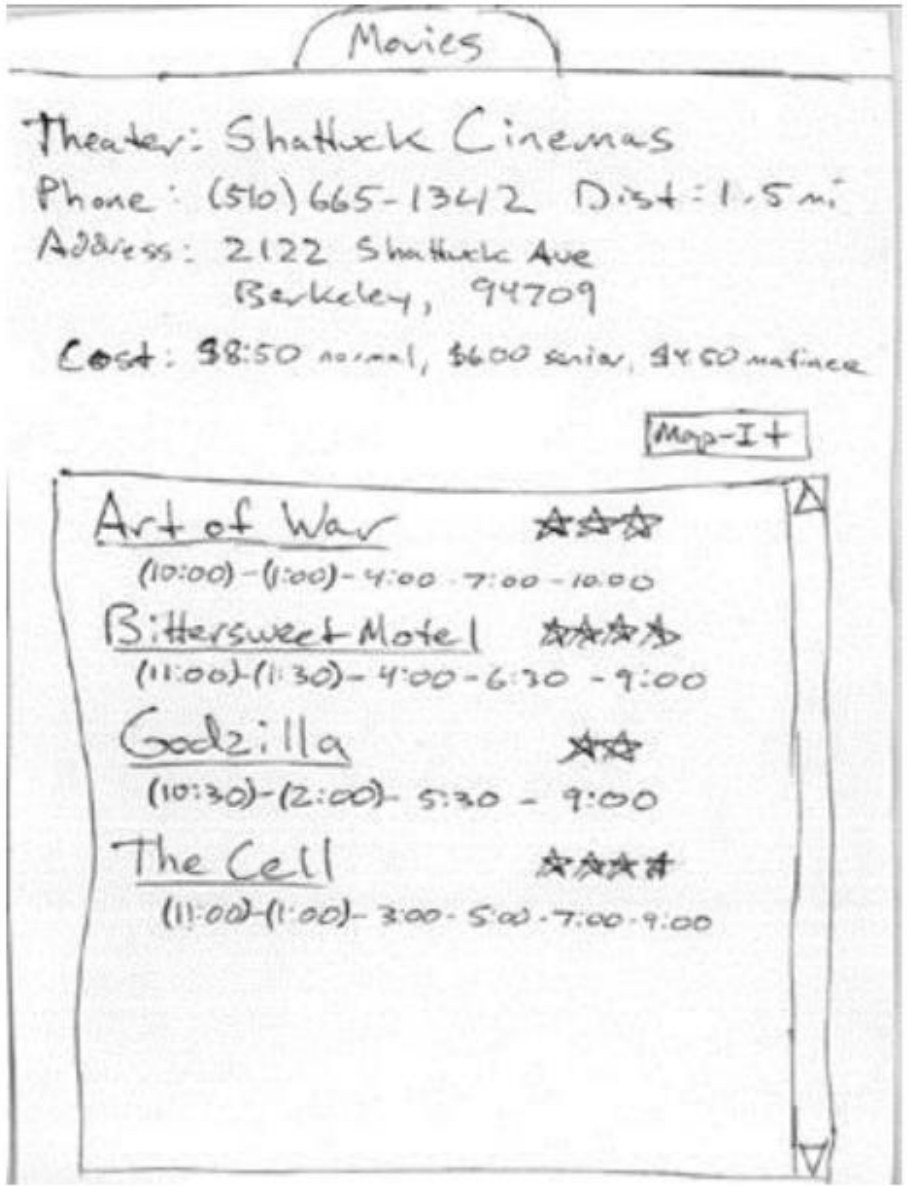
- Uses a medium which is unlike the final medium, e.g. paper, cardboard

- Is quick, cheap and easily changed

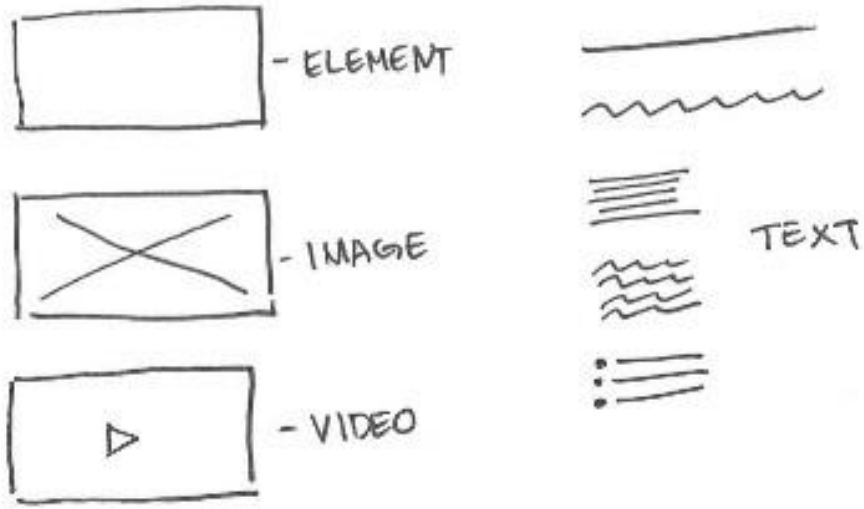
- Examples:
  - sketches of screens, task sequences, etc
  - ‘Post-it’ notes
  - storyboards

Sketching

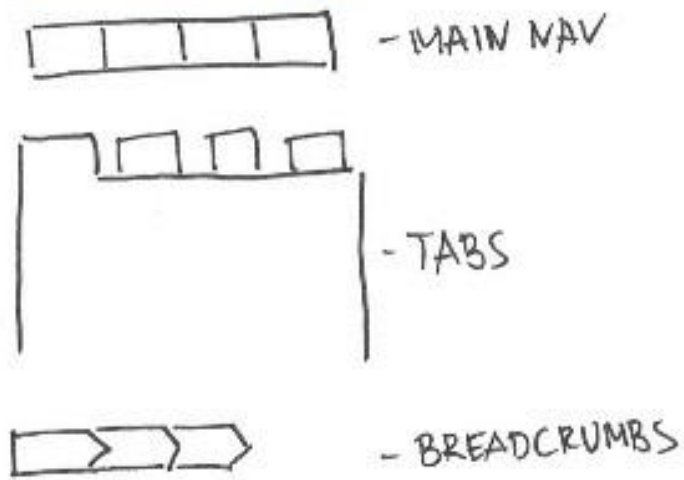
- Sketches are hand drawings on a piece of paper or in a digital tool that gives you a basic representation of your concept.
-



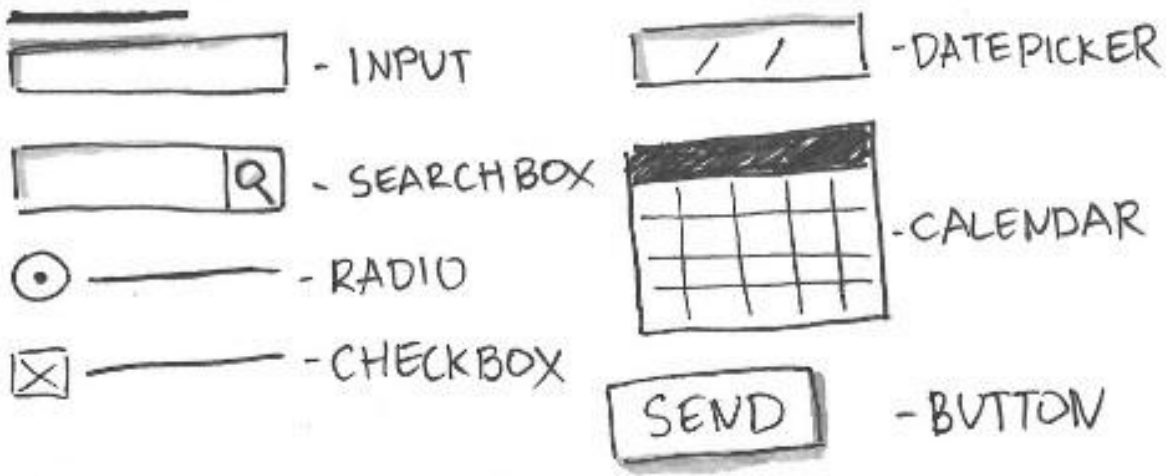
CONTENT:



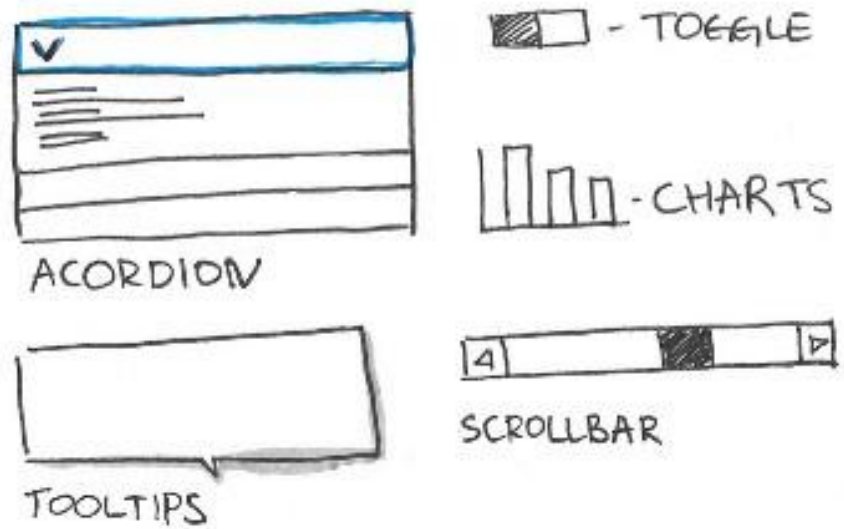
NAVIGATION:



FORMS:



OTHER:



uxCollective

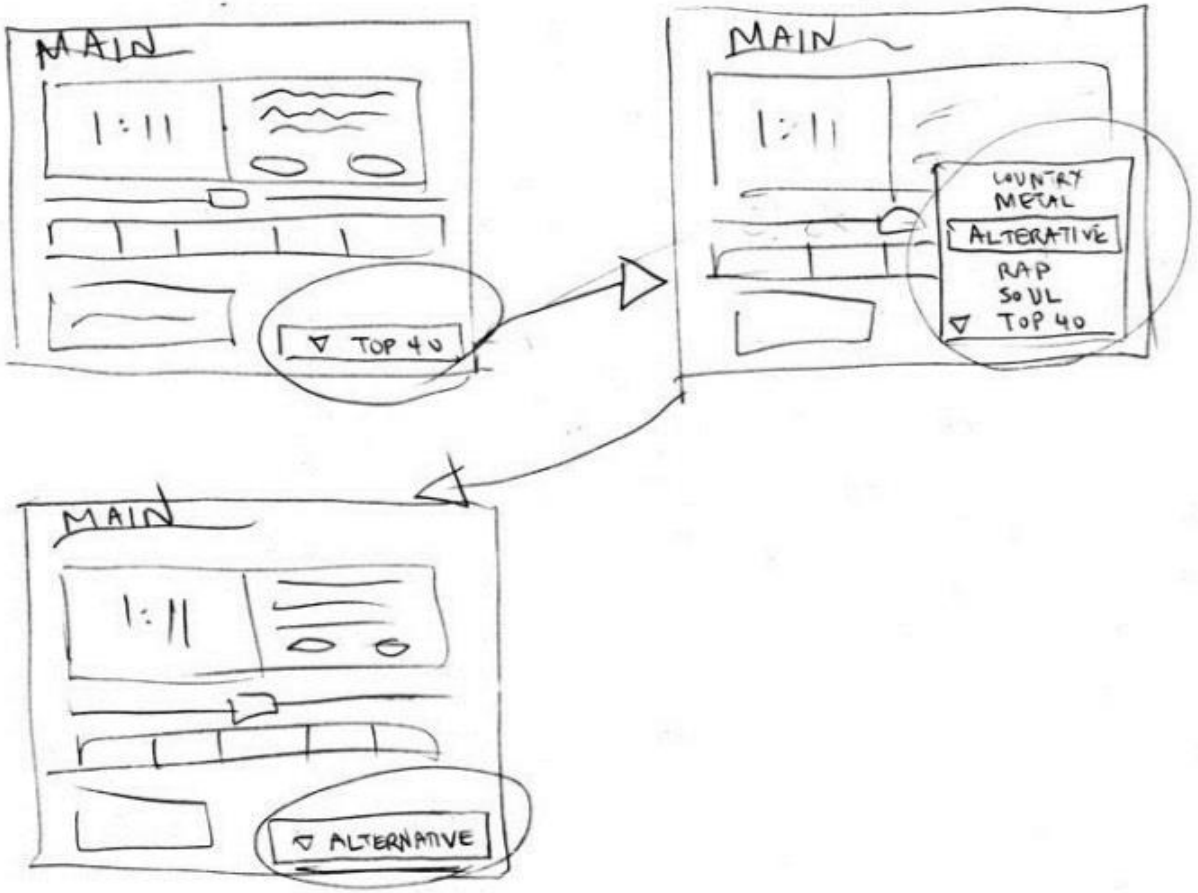
Sketching

When?

- Sketches can be very helpful during the conceptualization and initial visualization phase of your design process.
- People are visual learners, and visuals can explain ideas better than words.

Storyboards

- Often used with scenarios, bringing more detail, and a chance to role play
- It is a series of sketches showing how a user might progress through a task using the device
- Used early in design



Storyboards

- Storyboards are:
  - a low fidelity visual representation where
  - steps or actions represented by panels, like a comic book
- Goals are to
  - flesh out the scenarios in an interaction design
  - effectively communicate with users or stakeholders

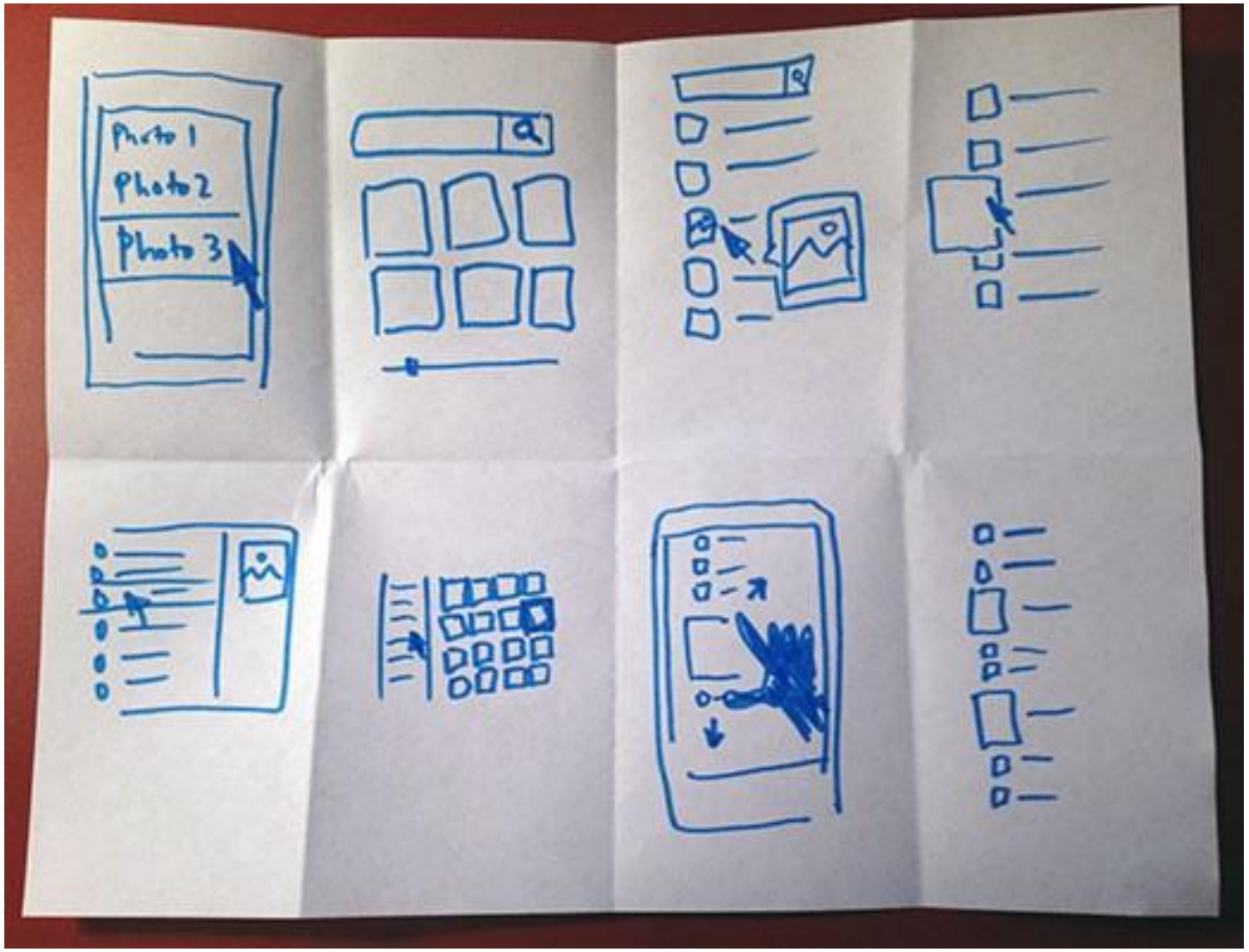
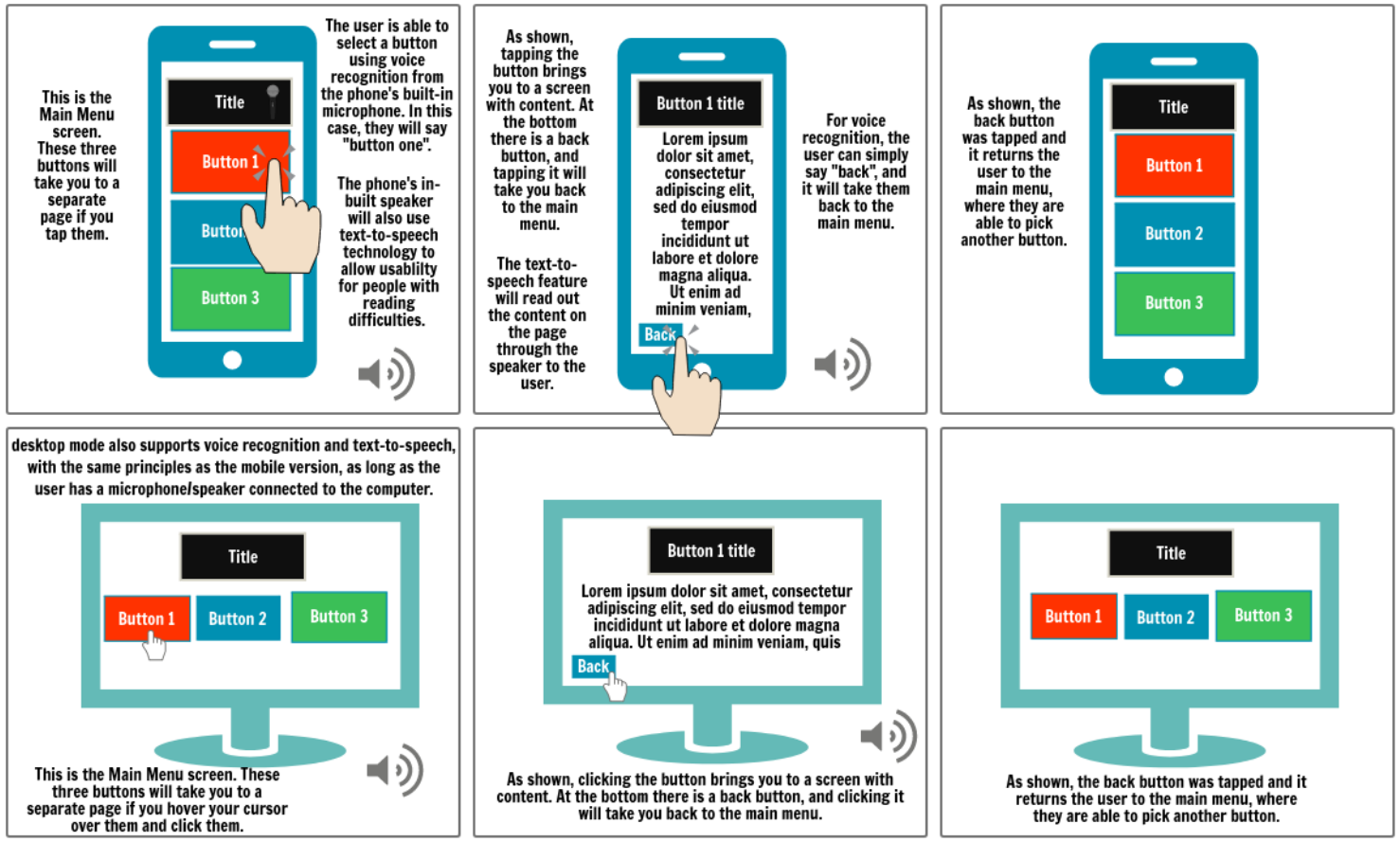


Photo 1
Photo 2
Photo 3

- Storyboard for a website
- Sequence of pages
  - based on clicks
- Explanations / annotations



From : *Storyboardthat*

Using Office Supplies

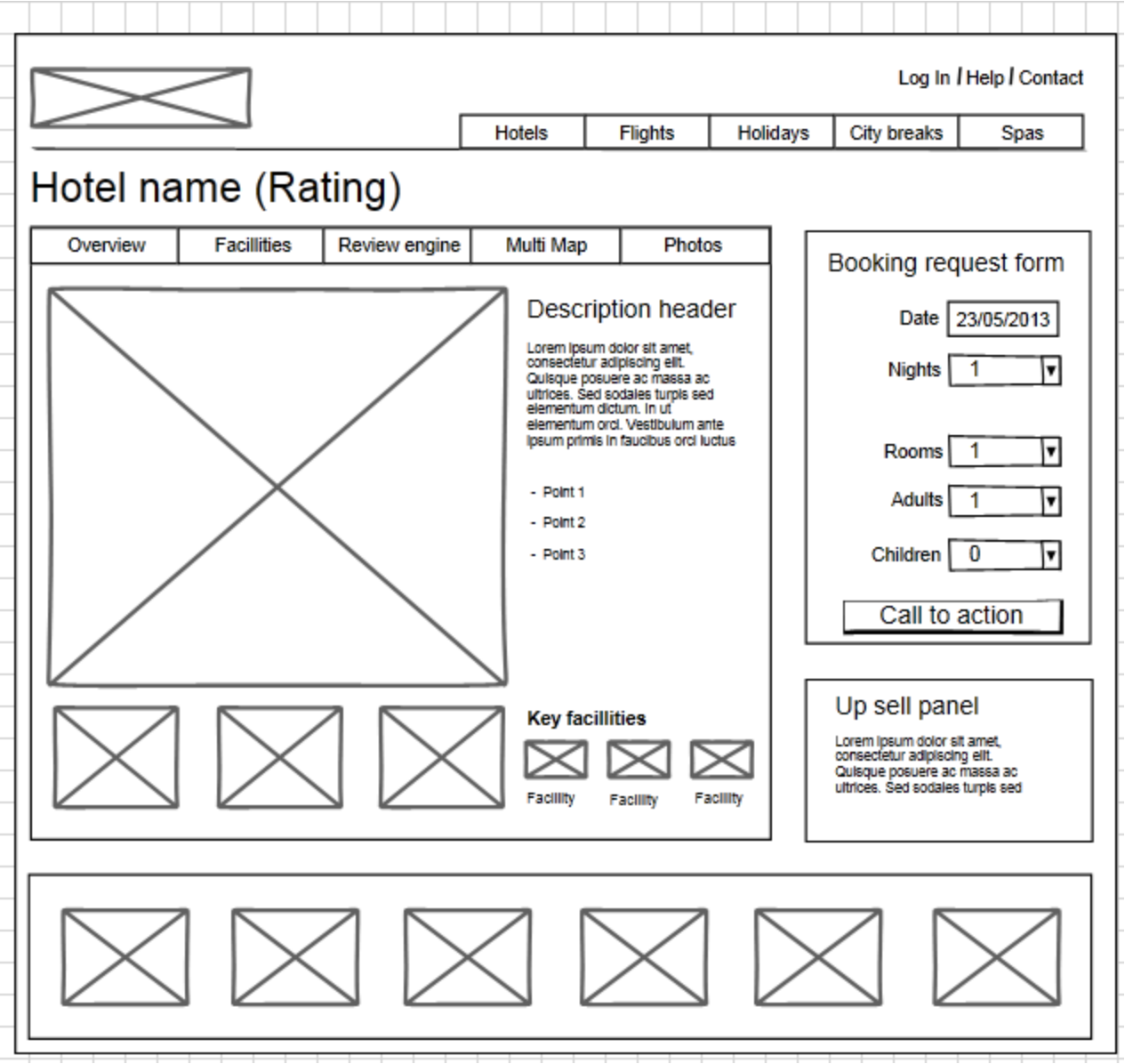
- Index cards, post-its
  - Index cards (3 X 5 inches)
  - Each card represents one screen
  - Often used in website development

Using Office Supplies

- Post-its, index cards
  - Can represent one screen, one page
  - Color coded
  - Draw on them
  - Group them
  - Put them on a wall or whiteboard, connect them with string or line

Wireframes

- Wireframes are low-fide design artifacts that represent only essential elements of UI
- Wireframes act as skeletons for your desig — they depict the basic and become a blueprint your product.



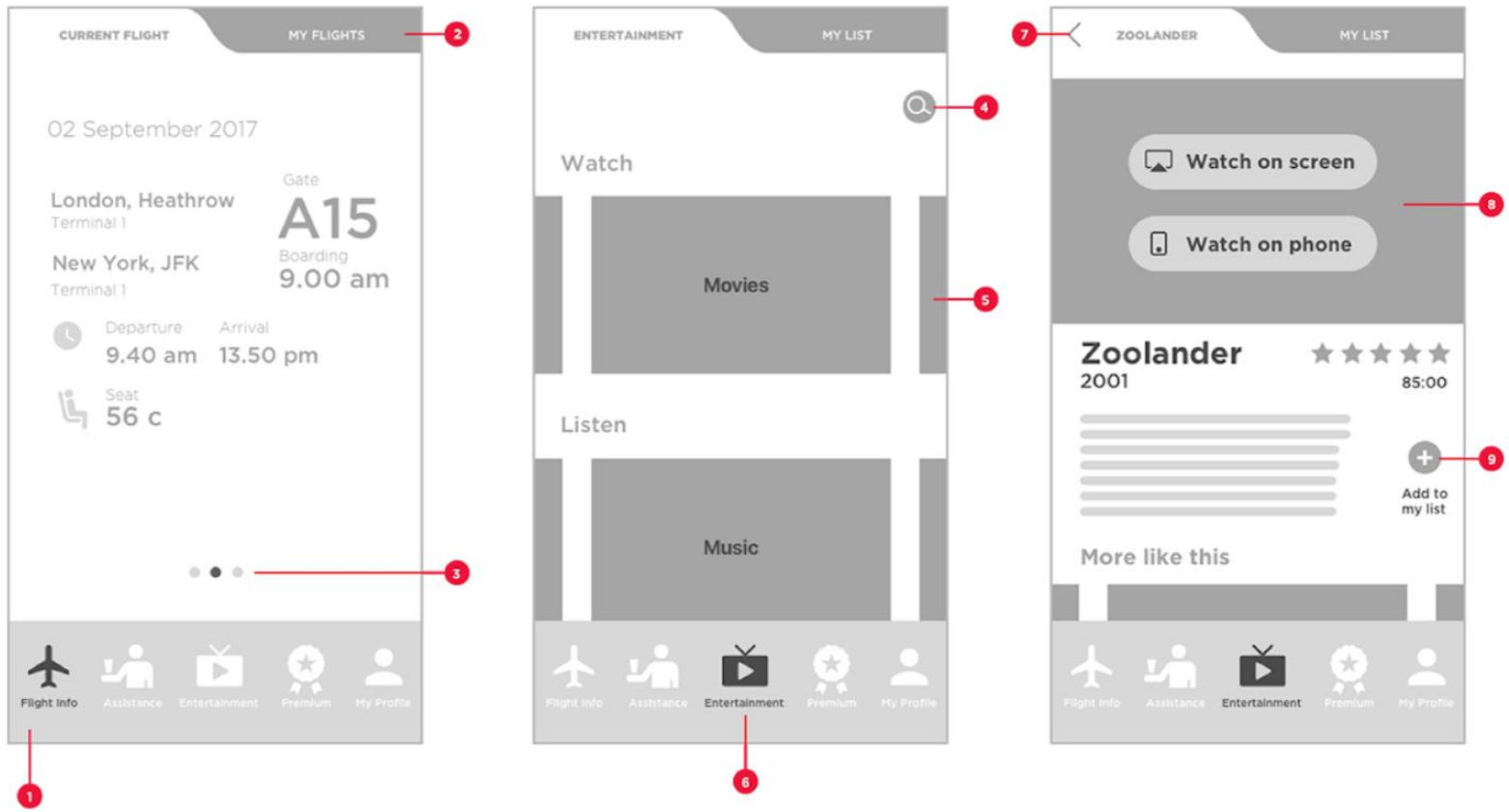
Wireframes

When?

- Wireframes are most relevant in the initial stages of the product design process. Wireframes are useful for:
  - Evaluating the structure of individual pages/screens;
  - Understanding how related screens/pages work together (from a user perspective);
  - Preparing a detailed project requirements document (wireframes can serve as an excellent reference).

How?

- Similar to sketches, wireframes can be created using pen & paper. When it comes to digital tools, Balsamiq perhaps is the most useful tool.



1. Global Navigation. It displays the five main features of the app
2. Secondary navigation. The user will access to the list of the upcoming and previous flights
3. By swiping on left/right the user will see boarding pass and flight map
4. Search icon. By clicking on it, the search bar will appear along with the keyboard
5. Carousel. It will display the list by sliding left/right
6. Different colour of the section icon, once selected
7. Backward arrow to navigate back
8. Two different options for watching: on screen and on phone
9. Possiblity to add to My List. The user can watch it later by accessing into the section My List.

High-fidelity prototyping

- Uses materials that you would expect to be in the final product.
- Prototype looks more like the final system than a low-fidelity version.
- For a high-fidelity software prototype common environments include Macromedia Director, Visual Basic, and Smalltalk.
- Danger that users think they have a full system.

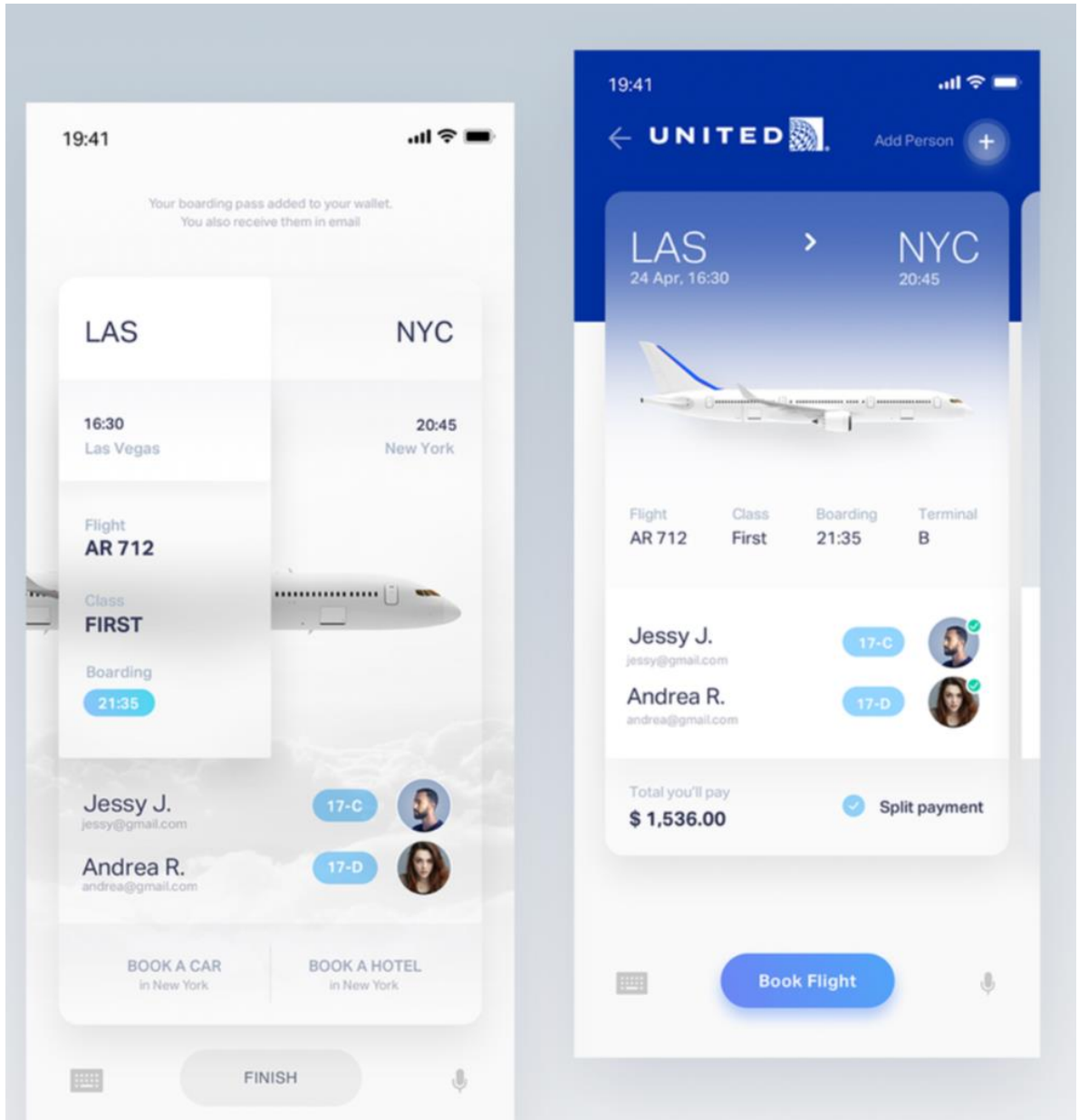
High-fidelity Prototyping

- Benefits
  - More realistic
  - Closer to final product
    - Good for developers and users
  - Can collect metrics
- Limitations
  - More expensive, less rapid
  - Reluctance to change

Mockups

- Mockups are a mid- or high-fidelity visualization of your design.
- Mockups deliver the visual look of the product design, and they are great for evaluating the look and feel of your design.

Mockup


19:41
Your boarding pass added to your wallet.
You also receive them in email
LAS
NYC
16:30
Las Vegas
20:45
New York
Flight
AR 712
Class
FIRST
Boarding
21:35
Jessy J.
jessy@gmail.com
17-C
Andrea R.
andrea@gmail.com
17-D
BOOK A CAR
in New York
BOOK A HOTEL
in New York
FINISH
19:41
UNITED
Add Person
LAS
24 Apr, 16:30
NYC
20:45
Flight
AR 712
Class
First
Boarding
21:35
Terminal
B
Jessy J.
jessy@gmail.com
17-C
Andrea R.
andrea@gmail.com
17-D
Total you'll pay
$ 1,536.00
Split payment
Book Flight

Mockup

Mockups are useful when a team wants to:

- ***Evaluate visual design decisions.*** See how colors, typography, and imagery work together.
- ***Experiment with various styles.*** Designers can try different color combinations and see what color scheme works best for users.
- ***Evaluate the visual consistency of design.*** Ensure that all screens in your product look like parts of the whole product, rather than a collection of individual screens.
- ***Evaluate the accessibility of your design***. Your design should allow users of all abilities to navigate, understand, and use your product. Focus on color contrast. Read W3C recommendations on how to select proper contrast ratios for body text and image text.
- ***Present the user interface to stakeholders.*** Sketches and wireframes often require clarification, while mockups are much easier to understand for people.

Mockup

When?

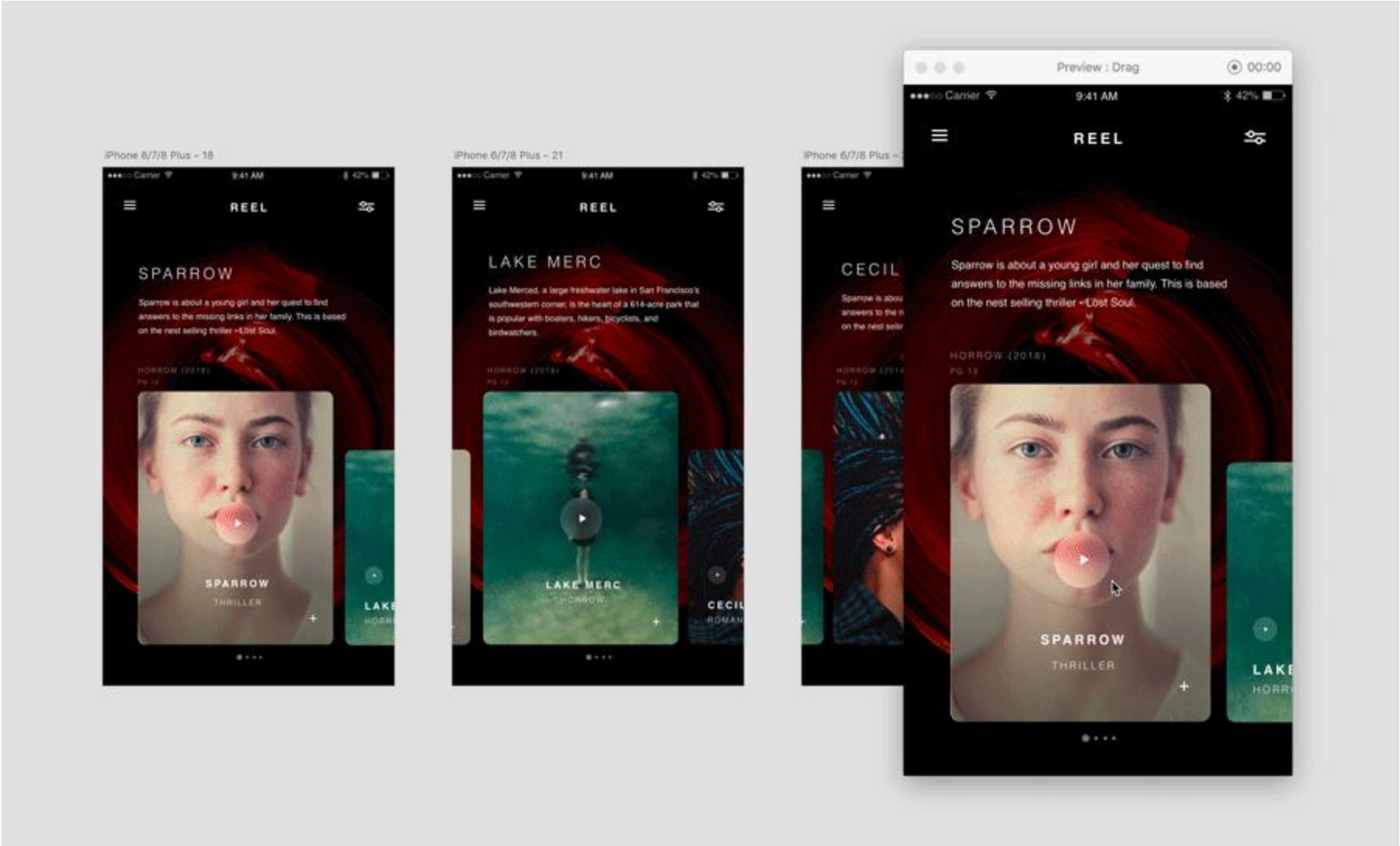
- Mockups can be very helpful during the visual design phase of design process.
- Mockups can be used both when designing a new product and redesign an existing product.

How?

- Mockups can be created in Photoshop, Sketch, Figma, and many other design tools.

Prototyping

- A prototype is a working model of an app/webpage.
- Prototypes allow designers to simulate user interaction.
- Unlike all other artifacts, prototypes are always interactive.
- The goal of a prototype is to simulate the interaction between the user and the interface.


iPhone 6/7/8 Plus – 18
REEL
SPARROW
Sparrow is about a young girl and her quest to find answers to the missing links in her family. This is based on the nest selling thriller «Lost Soul.
SPARROW
THRILLER
iPhone 6/7/8 Plus – 21
REEL
LAKE MERC
Lake Merced, a large freshwater lake in San Francisco's southwestern corner, is the heart of a 614-acre park that is popular with boaters, hikers, bicyclists, and birdwatchers.
LAKE MERC
Preview : Drag
00:00
Carrier
9:41 AM
42%
REEL
SPARROW
Sparrow is about a young girl and her quest to find answers to the missing links in her family. This is based on the nest selling thriller «Lost Soul.
HORROW (2018)
SPARROW
THRILLER

Compromises in prototyping

- All prototypes involve compromises
- Two common types of compromise
  - ‘horizontal’: provide a wide range of functions, but with little detail
  - ‘vertical’: provide a lot of detail for only a few functions

Possible Problems with Prototyping

- Pressure to enhance prototype to become delivered system
  - From client
  - From management
    - Both see code, see almost-working “system”

And then… Construction

- Taking the prototypes (or learning from them) and creating a final product
- Quality must be attended to: usability (of course), reliability, robustness, maintainability, integrity, portability, efficiency, etc